

HealthPlus Hospitals

Specialization: *Data Engineering*

Business Focus: *Healthcare Operations & Patient Management*

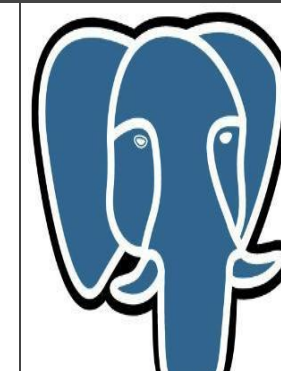
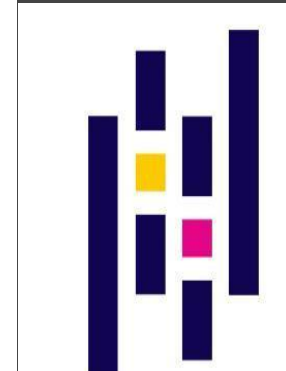
Tool: *Python (Pandas)*

HealthPlus Hospitals: Patient Data Modernization Project

Project Learning Opportunities

This project provides hands-on experience in using Pandas for real-world data engineering tasks. Students will learn how to load, explore, clean, validate, and transform healthcare data by handling missing values, removing duplicates, standardizing formats, and creating new business metrics. They will also learn how to merge datasets, prepare data for database storage, load cleaned data into PostgreSQL, and use Git and GitHub to manage and showcase their work as a complete Data Engineering project.

Tools and Technology to be Used



Case Study Overview

HealthPlus Hospitals is a rapidly expanding healthcare network with branches across Lagos, Abuja, Port Harcourt, and Ibadan.

The organization collects patient information from multiple hospital branches. However, due to manual entry and disconnected systems, the quality of the data has deteriorated over time.

Management wants a centralized, clean, and reliable patient database that can be used for operational reporting and business intelligence.

The Data Engineering team has been tasked with cleaning, transforming, validating, and preparing the data for migration into PostgreSQL.

HealthPlus
HOSPITALS —

Passion. Care. Commitment.

Health.
Lives.

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MAIN ENTRANCE



Diagnostic
Services



Specialist
Consultations



Pharmacy
Services

Case Study Overview

Introduction to the Business

HealthPlus Hospitals serves over 100,000 patients annually across Nigeria.

The hospital network provides:

- Outpatient services
- Emergency care
- Diagnostic services
- Specialist consultations
- Pharmacy services

Each hospital branch records patient information independently, causing inconsistencies across the organization.

Management now wants a unified healthcare data platform.



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Case Study Overview

Problem Statement

HealthPlus Hospitals collects patient data across multiple branches, but inconsistent data entry practices have resulted in missing values, duplicate records, invalid contact details, inconsistent formatting, and incorrect data types. These data quality issues make reporting unreliable and hinder effective decision-making. **The Data Engineering team has been tasked with cleaning, transforming, and validating the data before migrating it into a centralized PostgreSQL database.**



Rationale for the Project

Improved Data Quality

Accurate patient data improves operational efficiency.

Enable Better Reporting

Hospital executives need reliable reports for decision-making.

Support Future Analytics

Clean data creates a foundation for:

- Patient retention analysis
- Disease trend analysis
- Revenue reporting
- Resource planning

Centralize Health Records

Standardized data enables a single source of truth.



Table Overview

Patients Table

Stores:

- Patient ID
- Full Name
- Gender
- Age
- Phone Number
- Email
- City
- Registration Date

Appointments Table

Stores:

- Appointment ID
- Patient ID
- Doctor ID
- Appointment Date
- Appointment Status

Doctors' Table

Stores:

- Doctor ID
- Doctor Name
- Specialty
- Branch

Billing Table

Stores:

- Invoice ID
- Patient ID
- Amount Charged
- Amount Paid
- Payment Status



Project Workflow



Phase 1

Data Ingestion and Exploration

Load the raw healthcare datasets into Pandas, explore the data structure, and identify missing values, duplicates, inconsistent formats, and data quality issues.

Phase 2

Data Cleaning and Transformation

Clean and standardize the data by handling missing values, removing duplicates, correcting data types, formatting dates and text fields, creating new business metrics, and validating data quality.

Phase 3

Database Integration and Reporting

Merge the cleaned datasets, load them into PostgreSQL, perform business analysis, generate insights, document the project, and publish the completed solution to GitHub.

**READY TO
DELVE IN?**

